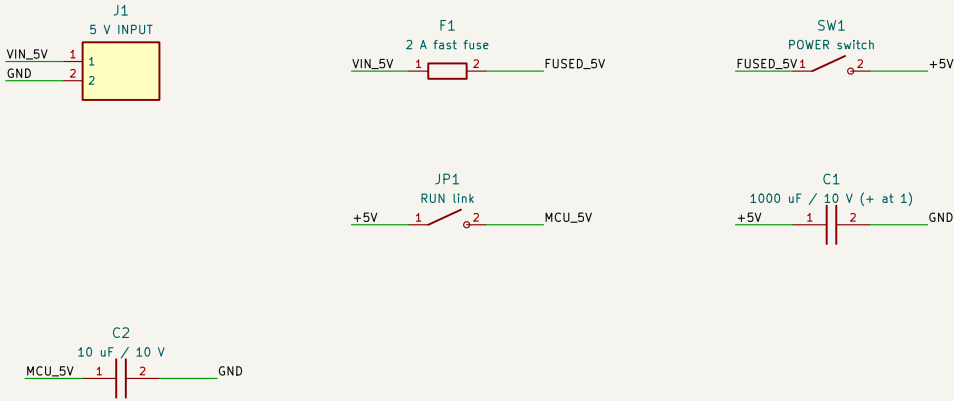


SOIL / VALVE – Rev A | MODULE–LEVEL BUILD SCHEMATIC

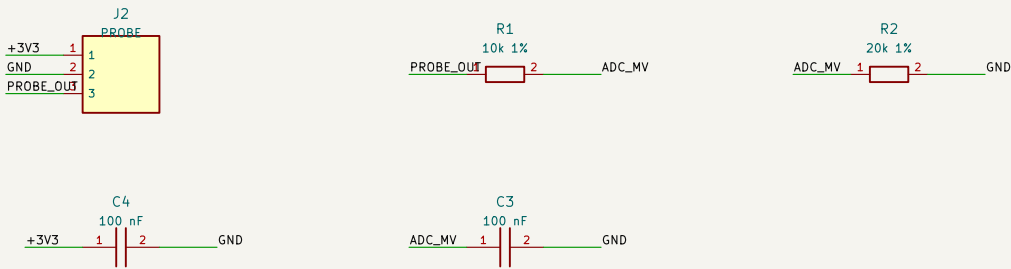
01 / POWER – regulated 5.0 V, 3 A SELV supply; star ground at J1



USB programming: SW1 OFF + JP1 REMOVED. Disconnect USB before restoring RUN.

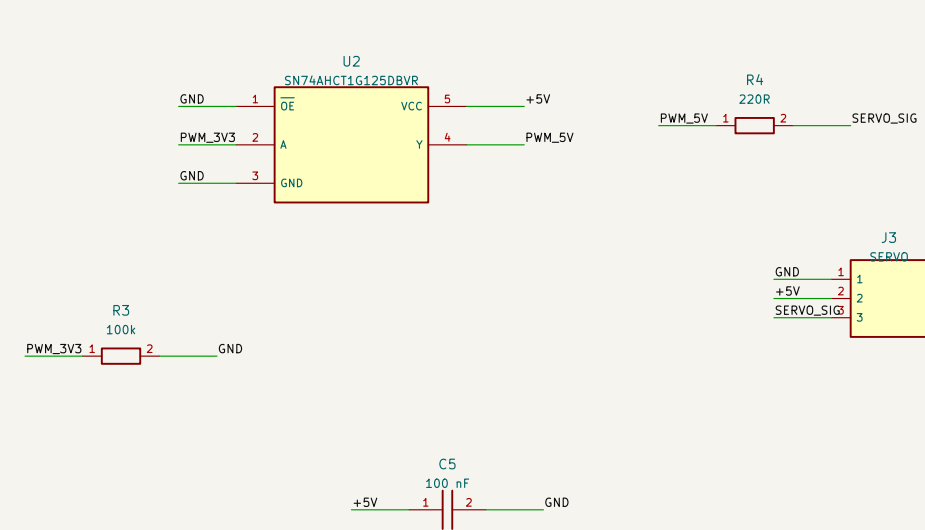
Input has no reverse–polarity protection; verify J1 polarity before connection.

02 / MOISTURE – SEN0193 powered from 3V3



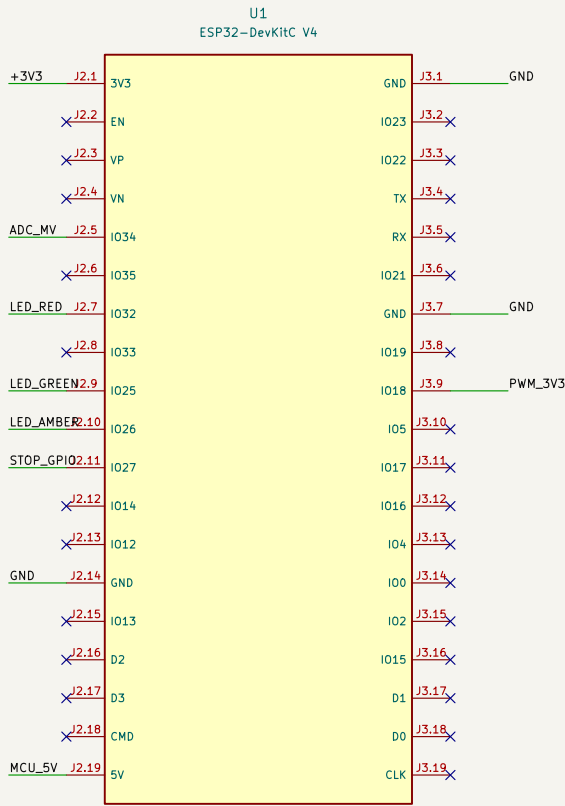
$V_{adc} = 0.667 \times V_{probe}$. Calibrate at ADC node with this divider installed.

03 / SERVO – HS–311; external low–pressure valve linkage



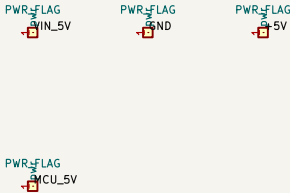
Servo current returns directly to J1 GND. Do not route through DevKit headers.

04 / ESP32 DevKitC V4 – ESP32–WROOM–32E

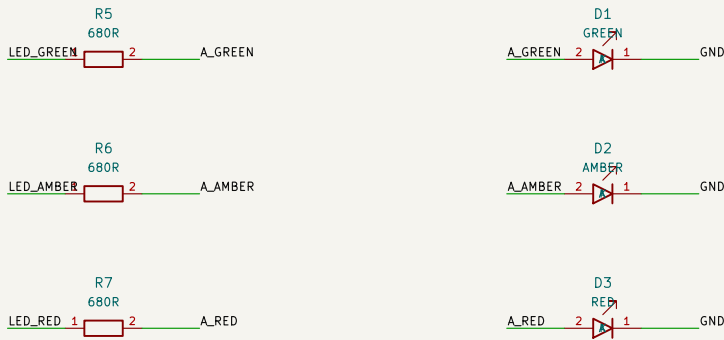


All unconnected module header pins intentionally unused.

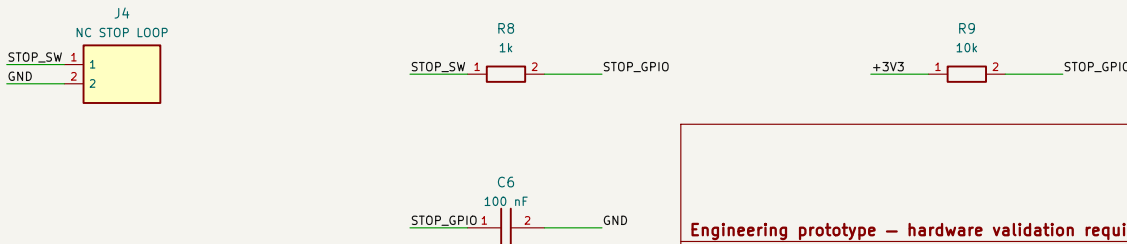
POWER SOURCE
DECLARATIONS



05 / PANEL LEDs – 5 mm diffused, active HIGH



06 / PHYSICAL STOP – firmware interlock, not a safety–rated E–stop



STOP latches a firmware fault.
No guaranteed closure on power loss.

Engineering prototype – hardware validation required

Sheet: /
File: soilValve.kicad_sch

Title: ESP32 Soil Moisture / Servo Valve

Size: A3 Date: 2026–09–25

KiCad E.D.A. 10.0.6

Rev: A

Id: 1/1